CSE558 DATA SCIENCE - LECTURE 12

The Bound at Work

Supratim IIIT-Delhi Monsoon 2026

WHERE WE STOPPED

The coin is resolved - now what?

$$Pr(X \geq a) \leq \frac{E[e^{tX}]}{e^{at}}$$

Q1 What bound did Chernoff give for P(S >= 50) on our coin, and at what value of t?

Q2 That t = ln(2.5) worked for THIS coin. Was it a lucky guess, or is there a general recipe?

WHAT WE'VE BUILT

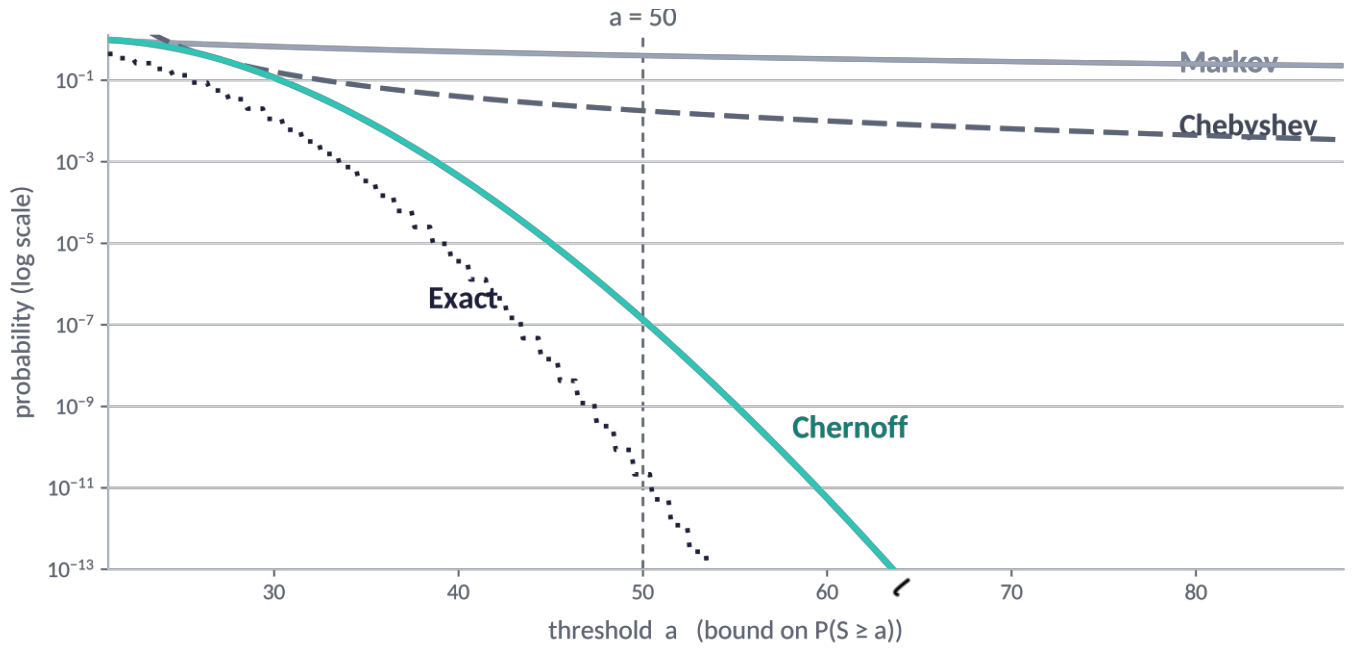
Three tools, one trick, repeated

a biased coin where up 0.2 it lands head.

Pr(# heads ≥ 50)

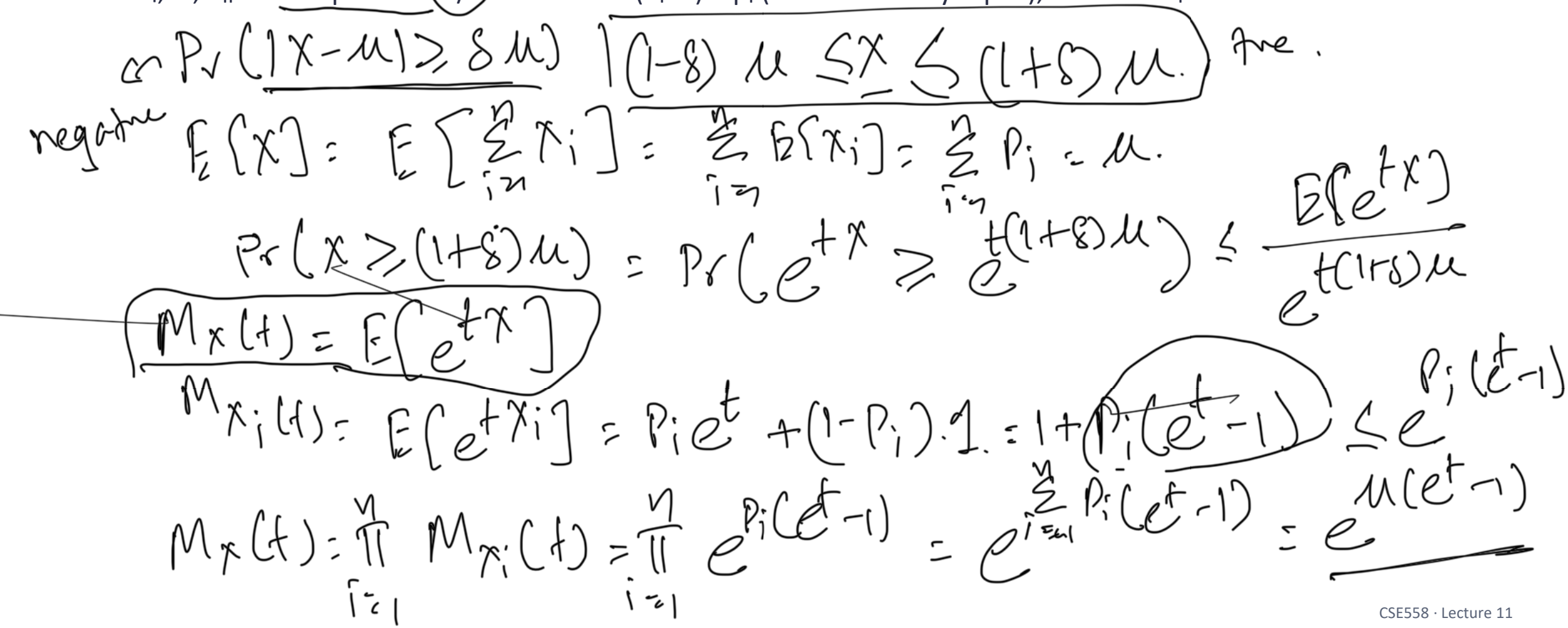
	Markov	Chebyshev	Chernoff
Needs	E[X]	E[X], Var[X]	every moment (via M_X(t))
P(S>=a): Tail shrinks	1/a	1/a^2	exponentially in a
Our coin: P(S>=50)	0.4	0.018	**1.35 x 10^-7**

How unlikely was 50 heads, really?



The Chernoff bound on a sum of Poisson trials

Let $X_1, \ldots, X_n$ be independent 0/1 trials with $P(X_i=1) = p_i$ (not necessarily equal), and $X = \Sigma X_i$.



The Chernoff bound on a sum of Poisson trials

Let $X_1, \ldots, X_n$ be independent 0/1 trials with $P(X_i=1) = p_i$ (not necessarily equal), and $X = \Sigma X_i$.

$$\Pr(X \geq (1+\delta)\mu) \leq \frac{e^{\mu(e^t-1)}}{e^{t(1+\delta)\mu}}$$

Let $t = \ln(1+\delta)$

$\Rightarrow e^t = 1+\delta$.

$$= \left(\frac{e^{1+\delta-1}}{(1+\delta)^{1+\delta}}\right)^{\mu} = \left(\frac{e^{\delta}}{(1+\delta)^{1+\delta}}\right)^{\mu}$$

The Chernoff bound on a sum of Poisson trials

Let $X_1, \ldots, X_n$ be independent 0/1 trials with $P(X_i=1) = p_i$ (not necessarily equal), and $X = \Sigma X_i$.

$$\Pr(X \le (1-\delta)\mu) = \Pr(e^{tX} \ge e^{t(1-\delta)\mu})$$

$$= \frac{e^{\mu(e^t-1)}}{e^{t(1-\delta)\mu}} = \left(\frac{e^{e^t-1-t}}{(1-\delta)^{1-\delta}}\right)^{\mu}$$

$$= \left(\frac{e^{-\delta}}{(1-\delta)^{1-\delta}}\right)^{\mu}$$

Let $t = \ln(1-\delta)$

$e^t = 1-\delta$

YOUR TURN

Is a server about to overload?

1,000 requests arrive and each, independently, is routed to one of 10 servers uniformly at random. Watch one specific server; let X be its load. What is the probability that the server gets a load more than 149 requests.

$E[X] = \frac{1000}{10} = 100$ μ.

$\delta = 0.5$

$\Pr(X \geq 150) \rightarrow (1+\delta)\mu$

$\leq \left(\frac{e^{\delta}}{(1+\delta)^{1+\delta}}\right)^{\mu}$

$= \left(\frac{e^{0.5}}{(1.5)^{1.5}}\right)^{100} = (0.9)^{100}$

EXIT TICKET

Is the judge under-accepting?

An online judge processes 500 independent submissions an hour. Historically each is Accepted with probability 0.3. Bound the probability that at most 90 submissions are accepted this hour.

Special Case

$X = \sum_{i=1}^{n} X_i$

Let $X_1, \ldots, X_n$ be independent -1/+1 trials with $P(X_i=-1) = P(X_i=+1) = ½$.

$M_{X_i}(t) = E[e^{tX_i}] = \frac{1}{2} \cdot e^t + \frac{1}{2} e^{-t} = \frac{1}{2} \sum_{i \geq 0} \left(\frac{t^{2i}}{(2i)!} \right) \times$

$= \sum_{i \geq 0} \frac{t^{2i}}{(2i)!} \leq \sum_{i \geq 0} \frac{(t^2/2)^i}{i!}$

$(2i)! \geq 2^i \cdot i!$

$e^t = 1 + t + \frac{t^2}{2!} + \frac{t^3}{3!} \ldots$

$e^{-t} = 1 - t + \frac{t^2}{2!} - \frac{t^3}{3!} + \ldots$

$= e^{t^2/2}$

$M_X(t) = e^{nt^2/2}$

$Pr(|X| \geq a)$,

$Pr(X \geq a) = Pr(e^{tX} \geq e^{ta}) \leq \frac{m_X(t)}{e^{ta}} = e^{nt^2/2 - ta}$

Special Case

Let $X_1, \ldots, X_n$ be independent -1/+1 trials with $P(X_i=-1) = P(X_i=+1) = ½$.

let $t = a/n$

$$Pr(X \geq a) \leq e^{\frac{a^2}{2n} - \frac{a^2}{n}} = e^{-\frac{a^2}{2n}}$$

$$Pr(X \leq -a) = Pr(e^{tX} \geq e^{-ta}) \leq \frac{e^{nt^2/2}}{e^{-ta}}$$

let $t = -a/n$

$$Pr(X \leq -a) \leq e^{-a^2/2n}$$

$$Pr(|X| \geq a) \leq e^{-a^2/2n} + e^{-a^2/2n}$$

$2e^{-a^2/2n}$.

Special Case

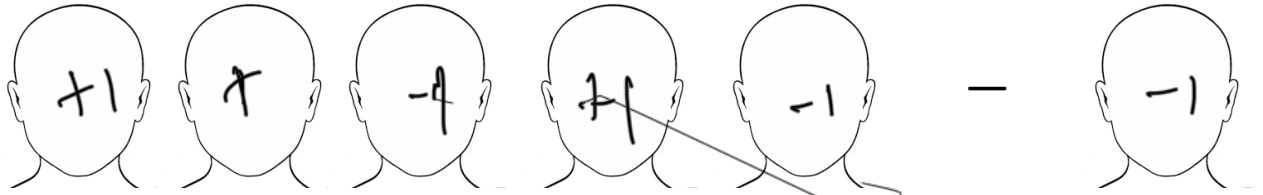
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PROBLEM

Set Balancing

Problem

You have m people, and each has some subset of n binary attributes such as smoker or not, over 40 or not, diabetic or not, and so on. You want to split the n people into two groups so that, for every attribute, the two groups end up with roughly equal counts of people who have it.



1	0	1	0	1	—	0
0	1	0	1	0	—	1
0	0	1	1	0	—	1
1	1	0	0	1	—	0

columns = m people

rows : n attributes

$A \in \{0,2\}^{n \times m}$

$b \in \{-1, \#\}^m$

a_1 :

a_2 :

a_n :

$a_i^T b = 0$

$a_i^T b$ is small

$\max_{i \in [n]} |a_i^T b|$ want small.

$\|Ab\|_\infty$

THE COST OF BEING EXACT

Why not just check every assignment?

There are 2^m possible vectors b in $\{-1, +1\}^m$. Checking one costs O(m^2) (computing Ab). Brute force checks all of them and keeps the best.

$$2^m = 2^{100} = 1.27 \times 10^{30}$$

Machine	Assignments/sec	Time for n = 100
A laptop core	~ 1 billion	~ 4 x 10^13 years
An exaflop supercomputer	~ 10^18	~ 40,000 years

Solution

$\Pr(\|A\vec{b}\|_\infty \geq k)$

b is random vector in $\{-1,+1\}^m$.

$$b_i := \begin{cases} +1 & \text{w.p. } 1/2 \\ -1 & \text{w.p. } 1/2 \end{cases}$$

Let i^{th} attribute is the row a_i of A. Among k a_i's.

if $k \leq \sqrt{4m\ln(n)}$ then $|a_i^T b| \leq \sqrt{4m\ln(n)}$

if $k \geq \sqrt{4m\ln(n)}$. let $z_i = a_i^T b = \sum_{j=1}^{m} a_{i,j}(b_j)$

$a_{i,j}(b_j)$: 0; $+1$ w.p. $1/2$; -1 w.p. $1/2$

$$\Pr(|z_i| \geq k) \leq 2e^{-2k^2/2m} = 2e^{-2\cdot 4m\ln(n)/2m} = \frac{2}{n^2}$$

The union bound

$$\Pr\left(\max_i |Z_i| \geq \gamma c\right) \leq n \cdot \frac{2}{n^2} = \frac{2}{n}$$

YOUR TURN

How balanced, for 100 volunteers?

Back to the trial: let 100 volunteers, 100 tracked attributes.